

Analysis of the Sales Price of Homes Sold in Ames, Iowa

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Understanding how housing prices behave is an important part of real estate analysis because sale values are shaped by many overlapping conditions, including the physical characteristics of a home, the quality of construction, the amount of living space, and the timing of when a property enters the market. Examining these relationships helps explain why similar homes may sell at very different prices and provides a foundation for later predictive modeling. This project focuses on the SalePrice variable and the housing attributes that may contribute to variation in value. To begin, descriptive statistical analysis was conducted on the training dataset before moving into more detailed regression-based evaluation in later stages. The Ames, Iowa housing dataset used for this analysis contains property records along with quantitative explanatory variables commonly used in housing price research. The variable definitions used by the Ames Housing data dictionary were published through Kaggle (Anna Montoya & DataCanary, 2016). The goal of the project is to establish a clear statistical understanding of how sale prices are distributed, how much variation exists across observations, and how the available housing characteristics may support future price prediction.

Before calculating descriptive measures, the training dataset was read into R and reviewed to confirm that the file imported correctly and that the variables were structured as expected. Figure 1 shows that the dataset contains 1,000 observations and 25 variables, which provides enough records to examine overall pricing patterns while supporting later comparisons across housing characteristics. The structure review also shows that all variables were imported as integer values. Although fields such as Id function primarily as identifiers and MSSubClass could also be interpreted as a categorical housing classification, the original data types were retained to preserve consistency with the provided dataset during the initial analysis. The output

also reveals missing values in selected variables, such as LotFrontage, which may become more relevant during later modeling steps. These fields were not adjusted at this stage, but they were noted for future reference in case they influence regression modeling or prediction accuracy later in the project.

Figure 1

Initial Import and Structural Review

```
> #-----
> #Portfolio Project Milestone 4
> #-----
> #Loading in the Tidyverse Library
> library(tidyverse)
> #Set Directory
> setwd("C:/Users/schwe/OneDrive/Documents/School/Data_Science_Foundations/data")
> #Read File into R
> trainingdata <- read.csv("MIS470HousingTraining(1000x25).csv", header = TRUE)
> #View Data & Structure to Confirm Data Imported Correctly
> glimpse(trainingdata)
Rows: 1,000
Columns: 25
$ Id <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25
$ MSSubclass <int> 60, 20, 60, 70, 60, 50, 20, 60, 50, 190, 20, 60, 20, 20, 20, 45, 20, 90
$ LotFrontage <int> 65, 80, 68, 60, 84, 85, 75, NA, 51, 50, 70, 85, NA, 91, NA, 51, NA, 72,
$ LotArea <int> 8450, 9600, 11250, 9550, 14260, 14115, 10084, 10382, 6120, 7420, 11200,
$ OverallQual <int> 7, 6, 7, 7, 8, 5, 8, 7, 7, 5, 5, 9, 5, 7, 6, 7, 6, 4, 5, 5, 8, 7, 8, 5,
$ OverallCond <int> 5, 8, 5, 5, 5, 5, 5, 6, 5, 6, 5, 5, 6, 5, 5, 8, 7, 5, 5, 6, 5, 7, 5, 7,
$ YearBuilt <int> 2003, 1976, 2001, 1915, 2000, 1993, 2004, 1973, 1931, 1939, 1965, 2005,
$ YearRemodAdd <int> 2003, 1976, 2002, 1970, 2000, 1995, 2005, 1973, 1950, 1950, 1965, 2006,
$ MasVnrArea <int> 196, 0, 162, 0, 350, 0, 186, 240, 0, 0, 0, 286, 0, 306, 212, 0, 180, 0,
$ TotalBsmtSF <int> 856, 1262, 920, 756, 1145, 796, 1686, 1107, 952, 991, 1040, 1175, 912, 1
$ GrLivArea <int> 1710, 1262, 1786, 1717, 2198, 1362, 1694, 2090, 1774, 1077, 1040, 2324,
$ FullBath <int> 2, 2, 2, 1, 2, 1, 2, 2, 2, 1, 1, 3, 1, 2, 1, 1, 2, 1, 1, 3, 1, 2, 1,
$ HalfBath <int> 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 0,
$ BedroomAbvGr <int> 3, 3, 3, 3, 4, 1, 3, 3, 2, 2, 3, 4, 2, 3, 2, 2, 2, 3, 3, 4, 3, 3, 3,
$ KitchenAbvGr <int> 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1,
$ TotRmsAbvGrd <int> 8, 6, 6, 7, 9, 5, 7, 7, 8, 5, 5, 11, 4, 7, 5, 5, 5, 6, 6, 6, 9, 6, 7, 6
$ Fireplaces <int> 0, 1, 1, 1, 1, 0, 1, 2, 2, 2, 0, 2, 0, 1, 1, 0, 1, 0, 0, 0, 1, 1, 1,
$ GarageYrBlt <int> 2003, 1976, 2001, 1998, 2000, 1993, 2004, 1973, 1931, 1939, 1965, 2005,
$ GarageCars <int> 2, 2, 2, 3, 3, 2, 2, 2, 2, 1, 1, 3, 1, 3, 1, 2, 2, 2, 2, 1, 3, 1, 2, 2,
$ GarageArea <int> 548, 460, 608, 642, 836, 480, 636, 484, 468, 205, 384, 736, 352, 840, 35
$ WoodDeckSF <int> 0, 298, 0, 0, 192, 40, 255, 235, 90, 0, 0, 147, 140, 160, 0, 48, 0, 0, 0
$ OpenPorchSF <int> 61, 0, 42, 35, 84, 30, 57, 204, 0, 4, 0, 21, 0, 33, 213, 112, 0, 0, 102
$ MoSold <int> 2, 5, 9, 2, 12, 10, 8, 11, 4, 1, 2, 7, 9, 8, 5, 7, 3, 10, 6, 5, 11, 6, 9
$ YrSold <int> 2008, 2007, 2008, 2006, 2008, 2009, 2007, 2009, 2008, 2008, 2008, 2006,
$ SalePrice <int> 208500, 181500, 223500, 140000, 250000, 143000, 307000, 200000, 129900,
```

Note. Output from `glimpse()` in R showing 1,000 observations, 25 variables, and initial variable data types after importing the training dataset.

Figure 2 summarizes the descriptive statistics for SalePrice and provides an initial view of how home values are distributed within the training dataset. Sale prices range from \$34,900 to \$755,000, indicating a wide spread between the lowest and highest observed values. This

suggests substantial variation across the properties. The mean sale price (\$182,284) is higher than the median (\$163,995), showing that the distribution is influenced by higher-priced homes that pull the average upward. This pattern suggests right skewness, where most homes are concentrated in the lower price ranges while fewer high-value properties extend the upper tail of the distribution (zyBooks, 2022). The standard deviation of \$80,291 further supports this spread by showing that sale prices vary considerably around the mean. Since the maximum is more than 3 standard deviations above the mean, it may also indicate the presence of high-value observations that behave as potential outliers. The first (\$130,000) and third quartiles (\$215,000) show that the middle 50 percent of home prices fall within a relatively broad range. This becomes useful when evaluating interval spacing for the histogram in the next stage of the analysis.

Figure 2

Descriptive Statistics for Sales Variable, Freedman-Diaconis Equation, & R Code for Histogram

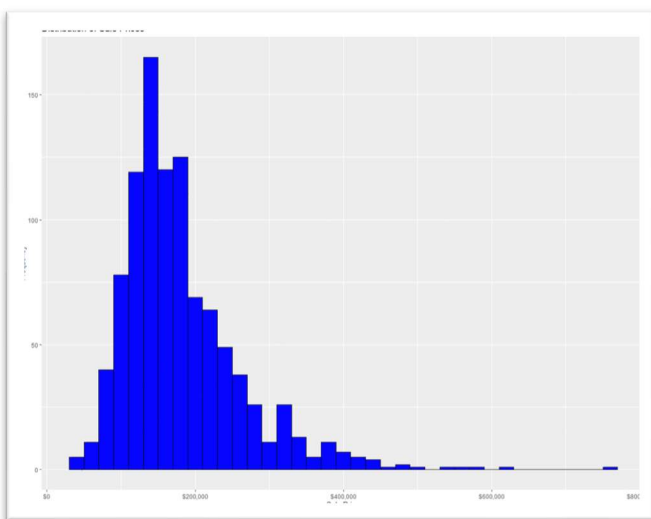
```
> #Descriptive Statistics for Sales Price Variable
> summary(trainingdata$SalePrice)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 34900 130000 163995 182284 215000 755000
> sd(trainingdata$SalePrice)
[1] 80291.22
> # Calculate the Freedman-Diaconis Bin width
> binwidth <- 2 * IQR(trainingdata$SalePrice) / (length(trainingdata$SalePrice)^(1/3))
> binwidth
[1] 17000
> #Create Histogram
> ggplot(trainingdata, aes(x = SalePrice)) +
+   geom_histogram(binwidth = 20000, fill = "blue", color = "black") +
+   labs(
+     title = "Distribution of Sale Prices",
+     x = "Sale Price",
+     y = "Frequency"
+   ) +
+   scale_x_continuous(labels = dollar)
> #Time & Date
> Sys.time()
[1] "2026-04-11 15:15:44 MDT"
```

Note. R output showing summary statistics for SalePrice, the Freedman-Diaconis bin width calculation, and the code used to generate the histogram in Figure 3.

Because the sale price distribution showed evidence of skewness and a wide spread between quartiles, the Freedman–Diaconis rule was used to estimate an appropriate histogram

bin width before visualizing the data. This method uses the interquartile range rather than standard deviation, which makes it less sensitive to extreme values and better suited for distributions that may contain outliers. Freedman and Diaconis originally proposed this approach as a way to improve histogram density estimation by selecting bin widths that better reflect the underlying shape of non-normal data (Jordan, 2026). Using the first and third quartiles from Figure 2, the calculated bin width was approximately \$17,000. This was rounded to \$20,000 to improve readability while remaining close to the formula-based result. Figure 3 shows the resulting histogram. Most of the sale prices cluster between approximately \$100,000 and \$250,000 while a smaller number of higher-priced homes extend the distribution to the right. This visual pattern reinforces the earlier indication of skewness and shows that although most observations fall within a moderate pricing range, several higher-value homes contribute to the extended upper tail of the distribution. These potential outlier observations were retained because they represent valid market variation rather than confirmed data errors, although their influence will remain important to monitor during later predictive modeling.

Figure 3
Histogram for Sales Price Frequency Distribution



Note. This figure shows the histogram of SalePrice using the Freedman–Diaconis method for bin width.

The initial analysis established a descriptive foundation for understanding how sale prices behave within the Ames housing training dataset. Reviewing the structure of the data confirmed that the file imported correctly and identified several variables that may require closer attention in later modeling stages, particularly where data type or missing values could influence results. The summary statistics showed substantial variation in home prices, with higher-value properties contributing to positive skewness across the distribution. Applying the Freedman–Diaconis rule provided a data-driven basis for selecting histogram intervals, and the resulting visualization reinforced the overall pricing pattern suggested by the numerical measures. These initial results provide a clear baseline for the next phase of the project, where housing attributes will be examined more directly through predictive modeling.

To continue the analysis, the training dataset was imported and reviewed as shown in Figure 1. The output confirmed that there were 460 observations across 25 variables, all of which matched the same structure and ordering of the training data. This consistency is important as it ensures that the datasets can be combined and interpreted without discrepancies. The review also indicated the presence of missing values, which were addressed later in the analysis.

The summary statistics on SalePrice for the testing dataset (Figure 4) show a distribution that closely aligns with the training dataset (Figure 2). The mean (\$177,958) is slightly lower than the training mean (\$182,284), and a similar pattern is observed in the median (\$161,750 vs. \$163,995). Since the testing standard deviation (\$77,569) is slightly lower than the training (\$80,291), it indicates that there is less variance in the testing data. Despite these differences, the overall distribution remains consistent across both datasets. This suggests that the testing dataset reflects similar underlying conditions, supporting its use for model validation.

Figure 4
Testing Dataset Import, Structure, and Descriptive Analysis

```

> #-----
> #Portfolio Project - Final
> #-----
> #Read Testing File into R
> testingdata <- read.csv("MIS470housingtesting(460x25).csv", header = TRUE)
> #View Data & Structure to Confirm Data Imported Correctly
> glimpse(testingdata)
Rows: 460
Columns: 25
$ Id <int> 1001, 1002, 1003, 1004, 1005, 1006, 1007, 1008, 1009, 1010, 1011, 1012, 1013, 1014, 10...
$ MSSubClass <int> 20, 30, 20, 90, 120, 80, 20, 160, 20, 50, 50, 90, 70, 30, 20, 60, 20, 120, 80, 120, 20...
$ LotFrontage <int> 74, 60, 75, NA, 43, 65, NA, 21, 43, 60, 115, 75, 55, 60, 60, 70, 73, NA, NA, 43, 60, 6...
$ LotArea <int> 10206, 5400, 11957, 11500, 3182, 8385, 12155, 2217, 12118, 6000, 21286, 9825, 10592, 7...
$ OverallQual <int> 3, 5, 8, 5, 7, 5, 6, 4, 7, 5, 5, 5, 6, 5, 6, 8, 7, 8, 7, 7, 4, 7, 5, 7, 8, 5, 5, 8, 5,...
$ OverallCond <int> 3, 6, 5, 6, 5, 8, 3, 4, 5, 5, 5, 5, 7, 4, 5, 6, 5, 5, 5, 5, 5, 5, 5, 6, 5, 5, 5, 5,...
$ YearBuilt <int> 1952, 1920, 2006, 1976, 2005, 1977, 1970, 1970, 1970, 2004, 1926, 1948, 1965, 1923, 1910, 19...
$ YearRemodAdd <int> 1952, 1950, 2006, 1976, 2006, 1977, 1970, 1970, 1970, 2005, 1950, 1950, 1965, 1996, 2006, 19...
$ MasVnrArea <int> 0, 0, 53, 164, 16, 220, 0, 0, 108, 0, 0, 0, 0, 0, 206, 0, 196, 0, 76, 145, 0, 84, 0, 1...
$ TotalBsmtSF <int> 0, 691, 1574, 1680, 1346, 985, 1657, 546, 1710, 1008, 720, 0, 602, 1022, 1082, 810, 15...
$ GrLivArea <int> 944, 691, 1574, 1680, 1504, 985, 1657, 1092, 1710, 1522, 1271, 1664, 1502, 1022, 1082,...
$ FullBath <int> 1, 1, 2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 1, 1, 1, 2, 2, 1, 2, 1, 2, 1, 2, 1, 1, 2, 1,...
$ HalfBath <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0,...
$ BedroomAbvGr <int> 2, 2, 3, 4, 1, 3, 3, 3, 3, 4, 4, 4, 3, 2, 2, 3, 3, 1, 3, 2, 2, 2, 2, 2, 3, 3, 3, 3,...
$ KitchenAbvGr <int> 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,...
$ TotRmsAbvGrd <int> 4, 4, 7, 8, 7, 6, 7, 6, 7, 7, 7, 8, 7, 4, 5, 6, 6, 4, 7, 6, 5, 6, 6, 7, 10, 5, 6, 7, 6...
$ Fireplaces <int> 0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 2, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 2, 1, 1,...
$ GarageYrBlt <int> 1956, 1920, 2006, 1976, 2005, 1977, 1970, 1970, 2004, NA, 1948, NA, 1923, 1956, 1948, ...
$ GarageCars <int> 2, 1, 3, 2, 2, 1, 2, 1, 2, 0, 1, 0, 1, 1, 1, 2, 2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 3, 1,...
$ GarageArea <int> 528, 216, 824, 528, 457, 328, 484, 286, 550, 0, 312, 0, 180, 280, 240, 528, 478, 565, ...
$ WoodDeckSF <int> 0, 0, 144, 0, 156, 210, 0, 238, 100, 0, 0, 0, 96, 0, 0, 0, 115, 63, 164, 142, 252, 105...
$ OpenPorchSF <int> 0, 20, 104, 0, 0, 0, 0, 48, 0, 0, 0, 0, 30, 130, 45, 66, 0, 0, 20, 64, 54, 0, 20, 7...
$ MoSold <int> 7, 1, 7, 6, 5, 11, 3, 8, 4, 6, 8, 5, 8, 6, 11, 11, 6, 8, 5, 4, 6, 7, 3, 5, 5, 3, 4, 4,...
$ YrSold <int> 2009, 2007, 2008, 2007, 2009, 2008, 2007, 2009, 2009, 2006, 2008, 2010, 2007, 2009, 20...
$ SalePrice <int> 82000, 86000, 232000, 136905, 181000, 149900, 163500, 88000, 240000, 102000, 135000, 1...
> #-----
> #Descriptive Statistics for Testing SalesPrice Variable
> summary(testingdata$SalePrice)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 52500 129000 161750 177958 206925 745000
> sd(testingdata$SalePrice)
[1] 77569

```

Note. This figure displays the structure of the testing dataset along with summary statistics and standard deviation of the SalePrice variable.

Figure 5 shows the R code used to create the histogram for the testing dataset, with the visualization shown in Figure 6. A bin width of \$20,000 was used instead of recalculating it with the Freedman-Diaconis rule to keep the histogram consistent with the one created for the training dataset. Using the same bin width makes it easier to compare the distributions and ensures that any differences seen are due to the data rather than changes in how the histogram is structured. The histogram in Figure 6 shows that most sale prices fall between roughly \$100,000 and \$250,000, with fewer homes at higher price levels. This pattern is very similar to the training dataset histogram (Figure 3), where the majority of homes also fall within a similar range and skewness. The testing dataset appears slightly more concentrated with fewer extreme high-value

observations, which aligns with the slightly lower variability observed in the summary statistics.

Similarly to the statistical summary, this suggests that the testing dataset follows the same distribution making it appropriate for evaluating the linear regression model.

Figure 5

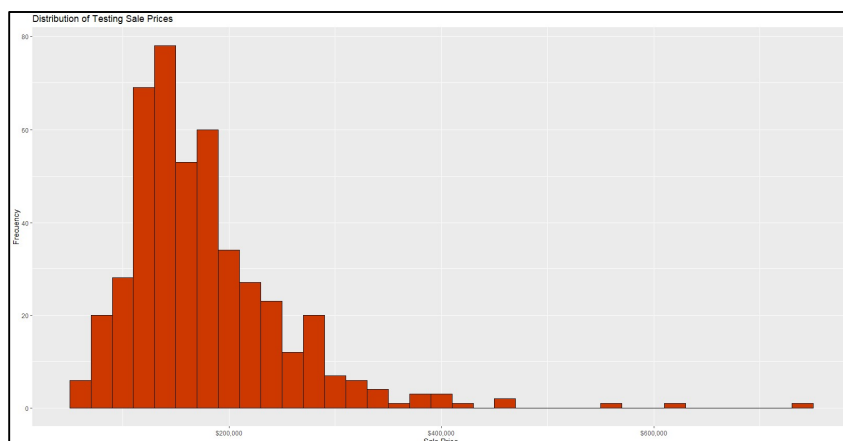
Code for Testing & Combined Histograms

```
> #-----
> #Create Testing SalePrice Histogram
> ggplot(testingdata, aes(x = SalePrice)) +
+   geom_histogram(binwidth = 20000, fill = "orangered3", color = "black") +
+   labs(
+     title = "Distribution of Testing Sale Prices",
+     x = "Sale Price",
+     y = "Frequency"
+   ) +
+   scale_x_continuous(labels = scales :: dollar)
> #Combining Training & Testing Data
> combineddata <- bind_rows(trainingdata, testingdata)
> #Create Combined SalePrice Histogram
> ggplot(combineddata, aes(x = SalePrice)) +
+   geom_histogram(binwidth = 20000, fill = "purple3", color = "black") +
+   labs(
+     title = "Distribution of Combined Sale Prices",
+     x = "Sale Price",
+     y = "Frequency"
+   ) +
+   scale_x_continuous(labels = scales :: dollar)
```

Note. R code used to generate the testing dataset histogram and the combined dataset histogram.

Figure 6

Histogram of Testing SalePrice



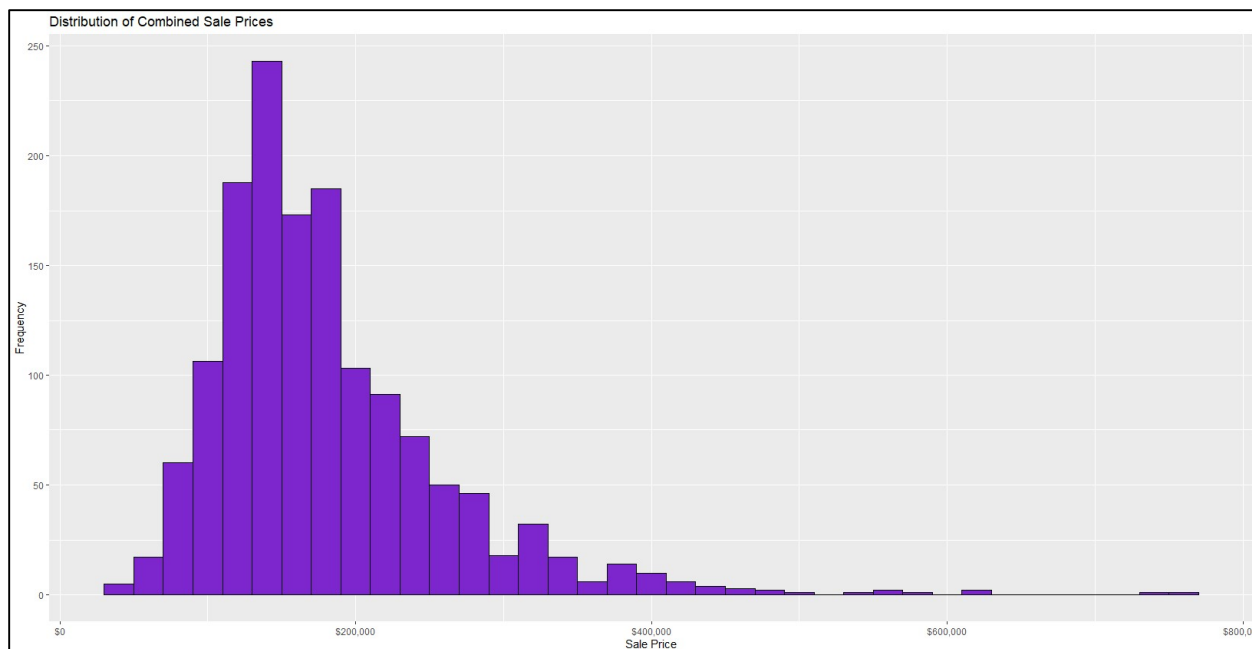
Note. Histogram showing the distribution of SalePrice for the testing dataset.

The training and testing datasets were combined into a single dataset labeled `combineddata`, as shown in Figure 5. The previously referenced `combine()` function is no longer supported within `dplyr`, so alternative methods were considered. Functions such as `vecr::vec_c()` were evaluated; however, they are primarily designed for combining vectors rather than full data frames. This could have led to unintended type coercion when working with structured tabular data. Join operations were also not appropriate in this context, as they are used to merge datasets based on key variables rather than stack observations. The `bind_rows()` function from the tidyverse was selected because it is specifically designed to combine data frames by appending rows while aligning variables by column name. This preserves the structure of the dataset (Wickham et al., 2023). While base R functions such as `rbind()` could also be used when datasets share identical structures, `bind_rows()` provides additional flexibility by aligning variables by name rather than position, making it a more robust option for combining tabular data.

The histogram for the combined dataset (Figure 7) shows a distribution that remains consistent with both the training (Figure 3) and testing datasets (Figure 6). The overall shape continues to exhibit a right-skewed pattern, with most sale prices concentrated between approximately \$100,000 and \$250,000 and fewer observations at higher price levels. Because the combined dataset includes all observations, the distribution appears slightly more spread out, with a more complete representation of higher-value homes. Despite this, the overall pattern does not change, reinforcing that both the training and testing datasets follow similar distributions and reflect the same underlying housing market conditions.

Figure 7

Histogram of Combined SalePrice



Note. Histogram showing the distribution of SalePrice for the combined dataset.

The linear regression model shown in Figure 8 was developed using the training dataset to establish a baseline for predicting sale prices. This model is then applied to the testing dataset for evaluation. The model demonstrates a strong overall fit, with an R-squared value of 0.847 and an adjusted R-squared of 0.842. This indicates that a large portion of the variation in sale prices is explained by the included variables. The model is also statistically significant overall ($p\text{-value} < 2.2e-16$), showing that the predictors collectively contribute meaningful information. Several variables are identified as significant contributors to sale price. Variables such as OverallQual, GrLivArea, TotalBsmtSF, and Fireplaces show strong positive relationships, meaning that higher-quality homes, larger living areas, and the number of fireplaces are associated with increased values. In contrast, variables such as KitchenAbvGr and BedroomAbvGr show negative relationships, which may reflect overlapping effects with other size-related variables included in the model. Understanding these relationships is important

because it provides context for how the model generates predictions when applied to the testing dataset.

Figure 8

Linear Regression Model Using Training Dataset

```
> #-----
> #Linear Regression Model for Training SalePrice
> traininglm <- lm(SalePrice ~ ., data=trainingdata)
> summary(traininglm)

Call:
lm(formula = SalePrice ~ ., data = trainingdata)

Residuals:
    Min       1Q   Median       3Q      Max
-378288  -17194   -1810   15486  235047

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.073e+06  1.864e+06  -0.576  0.56489
Id           4.910e+00  4.290e+00   1.144  0.25281
MSSubClass  -1.015e+02  3.716e+01  -2.732  0.00645 **
LotFrontage -7.293e+00  6.763e+01  -0.108  0.91416
LotArea      7.206e-01  1.480e-01   4.868  1.37e-06 ***
OverallQual  1.530e+04  1.634e+03   9.362 < 2e-16 ***
OverallCond  6.498e+03  1.536e+03   4.231  2.62e-05 ***
YearBuilt    4.289e+02  9.210e+01   4.657  3.80e-06 ***
YearRemodAdd 1.706e+02  9.408e+01   1.813  0.07016 .
MasVnrArea   3.279e+01  7.357e+00   4.458  9.56e-06 ***
TotalBsmtSF  2.581e+01  4.405e+00   5.858  7.01e-09 ***
GrLivArea    4.773e+01  6.250e+00   7.638  6.78e-14 ***
FullBath     -2.674e+03  3.782e+03  -0.707  0.47974
HalfBath      5.362e+02  3.379e+03   0.159  0.87394
BedroomAbvGr -1.497e+04  2.339e+03  -6.400  2.74e-10 ***
KitchenAbvGr -2.802e+04  7.166e+03  -3.911  0.00010 ***
TotRmsAbvGrd  9.040e+03  1.659e+03   5.448  6.93e-08 ***
Fireplaces    6.267e+03  2.359e+03   2.656  0.00807 **
GarageYrBlt   -9.373e+01  9.898e+01  -0.947  0.34393
GarageCars     5.174e+03  3.864e+03   1.339  0.18093
GarageArea     4.690e+01  1.331e+01   3.524  0.00045 ***
WoodDeckSF     2.016e+01  1.119e+01   1.802  0.07197 .
OpenPorchSF   -2.212e-01  2.136e+01  -0.010  0.99174
MoSold        -8.515e+02  4.709e+02  -1.808  0.07095 .
YrSold         2.068e+00  9.300e+02   0.002  0.99823

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33730 on 746 degrees of freedom
(229 observations deleted due to missingness)
Multiple R-squared:  0.8473,    Adjusted R-squared:  0.8423
F-statistic: 172.4 on 24 and 746 DF,  p-value: < 2.2e-16
```


Note. R output showing the removal of observations with missing values (NA), followed by the selection of the first 20 rows from the cleaned testing dataset for use in prediction.

Shown in Figure 10, the cleaned testing dataset was used to evaluate the performance of the linear regression model developed from the training data. This step is important because the training dataset is used to build the model, while the testing dataset is used to assess how well the model performs on new, unseen data. Evaluating the model on separate data helps reduce the risk of overfitting, where the model performs well on the training data but does not generalize effectively to new observations (Yadav, 2024). Using the first 20 observations from the cleaned testing dataset, predicted sale prices were generated and compared to the actual values. A data frame was then created to display the predicted and actual sale prices side by side, allowing for a direct comparison of model performance across individual observations.

Figure 10
Prediction of Sale Prices and Comparison to Actual Values

```
> #Predict Sales Price using first20cleantesting
> prediction <- predict(traininglm, newdata=first20cleantesting)
> #Comparing Actual vs Predicted values
> comparison <- data.frame(
+   Actual = first20cleantesting$SalePrice,
+   Predicted = prediction)
> comparison
  Actual Predicted
1   82000   42543.91
2   86000   68628.48
3  232000  271105.78
5  181000  238326.10
6  149900  132642.92
8   88000   66500.07
9  240000  250805.98
11 135000  108329.24
13 165000  154978.65
14   85000   81968.45
15 119200  135811.68
16 227000  211318.49
17 203000  220154.49
20 213490  218596.34
21 176000  145999.29
22 194000  211844.99
23   87000  110040.54
24 191000  222438.49
26 112500  110296.69
27 167500  163728.54
> #-----
> #Mean Absolute Error (MAE) & Mean Absolute Percentage Error (MAPE)
> mae <- mean(abs(comparison$Actual - comparison$Predicted))
> mae
[1] 20269.98
> mape <- mean(abs((comparison$Actual - comparison$Predicted) / comparison$Actual)) * 100
> mape
[1] 14.58857
> #Visualization for Actual vs Predicted values
> ggplot(comparison, aes(x = Actual, y = Predicted)) +
+   geom_point(size = 3, alpha = 0.7, color = "steelblue4") +
+   geom_abline(intercept = 0, slope = 1, linewidth = 1, color = "red") +
+   labs(
+     title = "Actual vs Predicted Sale Prices",
+     x = "Actual Sale Price",
+     y = "Predicted Sale Price"
+   ) +
+   scale_x_continuous(labels = scales::dollar) +
+   scale_y_continuous(labels = scales::dollar)
> #-----
> #Time & Date
> Sys.time()
[1] "2026-05-03 16:05:37 MDT"
```

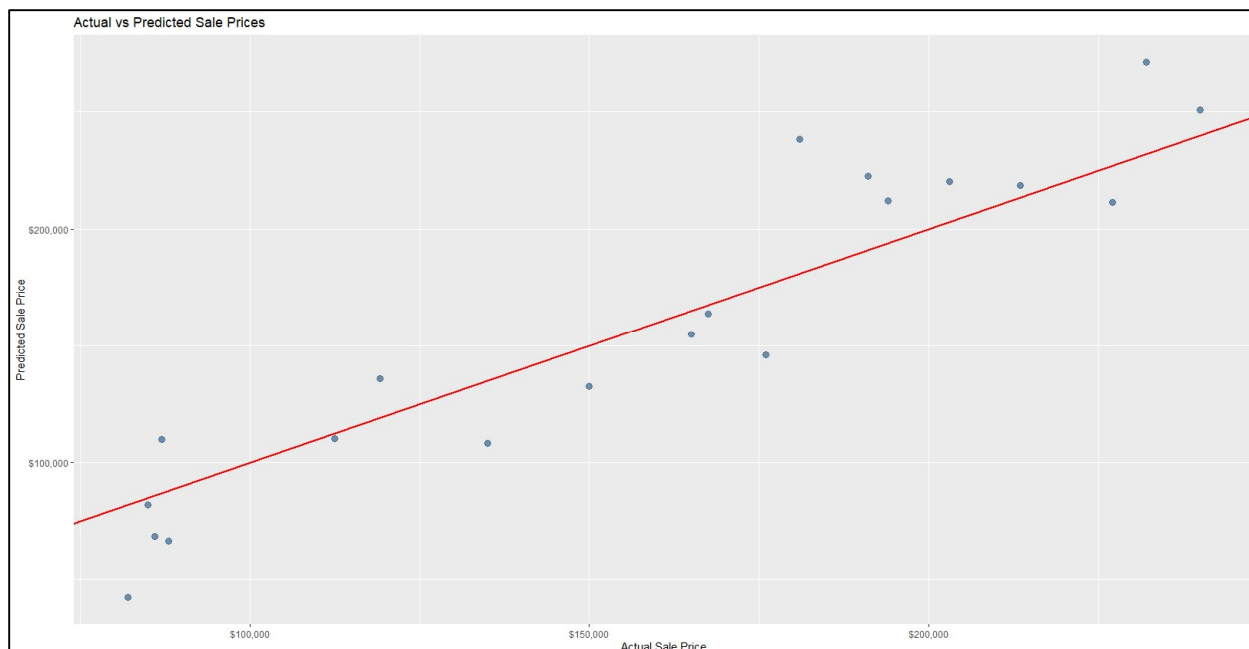
Note. This figure shows the predicted sale prices generated from the regression model, comparison of predicted and actual values, calculation of MAE and MAPE, and code used to create the actual versus predicted scatterplot.

A comparison of the predicted and actual sale prices shows that the model is generally able to estimate values within a reasonable range, although differences are present across individual observations. Some predicted values are close to the actual sale prices, while others show larger gaps, particularly for higher-priced homes. To quantify this difference, the mean absolute error (MAE) was calculated at approximately \$20,270, indicating the average deviation between predicted and actual values. The mean absolute percentage error (MAPE) was calculated at approximately 14.6%, providing a relative measure of prediction accuracy across different price levels. Relative to average home prices, these values indicate a moderate level of accuracy, with predictions generally falling within a reasonable range of the actual sale prices (zyBooks, 2022).

To further evaluate model performance, a scatterplot of actual versus predicted sale prices is shown in Figure 11. The red reference line represents perfect prediction, where predicted values would exactly match the actual sale prices. The plotted points generally follow an upward trend along this line, indicating that the model is capturing the overall relationship between the variables and sale price. However, there is visible dispersion around the line, showing that prediction errors are present across observations. This variation aligns with the MAE and MAPE values, reinforcing that while the model captures general pricing trends, it does not perfectly predict individual sale prices. Even so, the visualization supports the conclusion that the model provides a reasonable level of predictive accuracy with some variability across different price levels.

Figure 11

Actual vs. Predicted Sale Prices



Note. Scatterplot comparing actual and predicted sale prices for the testing dataset. The red reference line (slope = 1) represents perfect prediction, where predicted values equal actual values.

The analysis shows that sale prices in the Ames, Iowa housing dataset follow a consistent distribution across the training, testing, and combined datasets, with most homes concentrated in a similar price range and a smaller number of higher-value properties extending the upper end. The linear regression model demonstrated a strong fit, identifying key factors such as overall quality and living area as important drivers of sale price. When applied to the testing dataset, the model produced reasonably accurate predictions, with an average error of approximately \$20,270 and a relative error of about 14.6%. While some variation remained across individual observations, particularly among higher-priced homes, the results indicate that the model is effective at capturing overall pricing trends. These findings highlight the value of combining statistical analysis and predictive modeling to better understand housing markets and support more informed decision-making.

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